

Toyota e-Palette Concept

An autonomous truck that can be anything from a shop to an office to a restaurant and more

Over the years, CES has seen its fair share of amazing but totally unrealistic projects that have fed the hopes of people around the world only to never see the light of the day in the long run. That is what makes Toyota's e-Palette concept special – the company seems to have a well-thought out plan and partnerships in place to actually make it happen. The e-Palette, as a concept, is deceptively simple – it is a box with eight wheels. It can be fitted with chairs, tables, screens, shelves, kitchen counters, beds – you get the drift. There's no driving mechanism available so its safe to say that the e-Palette will be fully autonomous.

Basically, e-Palette is a blank slate for urban mobility, something that is essential from an automotive company during a time when the automobile industry is facing stiff competition from Silicon Valley companies in innovation. Imagine this, you order a meal from a restaurant, and instead of a meal being cooked and sent to you, the restaurant itself comes to you, all the while cooking the particular meal that you've ordered – thus reducing delivery time while



maintaining the quality of the food. Similar examples are aplenty in all areas.

Toyota is already collaborating with companies like Amazon, Mazda, Uber, Didi Chuxing and Pizza Hut to optimise the technology that can make its pods a reality, which are currently planned in three sizes. If and when this concept

does become real, it will provide an end-to-end solution for the self driving market that will make transportation a constant state of being instead of an activity that you undertake on a requirement basis. Get ready for a future of specialised pods that will bring customised services to you while you travel.

Omron Forpheus

Talking of robots, another popular robot this year at CES was the Omron Forpheus, a table-tennis tutor. While the robot itself isn't brand new and has been a standard feature at many tech shows over the last couple of years, this was the first time Omron attended CES and displayed its upgraded capabilities publicly. The robot has a host of sensors like a pair of cameras on either side to function like human eyes for tracking the ball up to 80 times per second, along with a central camera to observe the opponent's movements and facial expressions. Based on this data, Forpheus can judge



the opponent's skill level and adjust difficulty accordingly, with a goal to maximise a rally. This year, Forpheus has added certain capabilities – like the ability to serve, to predict a smash and to produce sounds. The robotic arm on the Forpheus is a re-purposed unit that used to be a packaging robot, has a five axis motor system that is used to hit the ball back. Paddle grip, ball hit location and arm position are all controlled to within 0.1mm. Omron's primary objective with this setup wasn't to sell a table tennis tutor to the world, but to showcase the company's robotics and AI capabilities.